0 or 1 Journal of Cards for Engineering Applied Mathematics IIB
Volume 1(2026), Issue 1, pages 1-12

Received: March 10, 2026; Accepted: October 10, 2026; Published: October 19, 2026

Engineering Applied Mathematics IIB for Two Players with Six Number of Jester Cards on Differential Calculus of Function of Several Variables–Level 2

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Abstract

This work presents 16 number of cards from applied mathematics IIB degree level mainly focused on differential calculus of function of several variables. The jester cards are A, B, C, D, E and F.

1 Introduction

At present, the decline in the reading community has reached a worrisome level. For many reasons, people have stopped sitting down and reading. For instance, wasting time on useless social media and in trivial places are among the many minor reasons. In 2025, Jenber proposed a solution to this through a major share company established under the title "Zero or One Trading Share Company." This share company believes that the problem mentioned above can be solved in the following ways. We found the solution included in this writing to be convincing (for more information, see the foundation document of 0 or 1 game [4]). The solution presented by this share company is briefly summarized below:

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(1) Making reading or studying entertaining, profitable, and rewarding. One of the methods for studying or acquiring new knowledge is understanding things by associating them with their equivalents. For example, an English word has many synonyms. Trying to memorize them by associating them is considered one of the best methods. However, to make this exercise unforgettable, doing it in the form of a game while having fun and discussing in groups makes it preferable.There are scientific studies conducted around studying English vocabulary by preparing cards (for example, see [1, 2, 3, 5, 6, 7, 10, 11, 12, 14]).

(2) Understanding things by associating them according to their types stimulates thinking, makes one inquisitive, sparks discussions with friends, and even drives one to search for reference books. And when done this way, it takes a much longer time to forget that thing.

(3) Furthermore, if this association is done enjoyably in the form of a game with friends, and if a bonus/reward is given to the winner to encourage knowledge and learning, the game will become popular.

(4) Developing various types of entertaining games to execute the solutions mentioned in 1-3 above.

In this paper, we have prepared cards for the 0 or 1 game. We have briefly described the number of cards we prepared as follows. Let the number of jester cards be 6 and the number of players be $k = 2$. Therefore, the total number of cards, $n = 8 \times 2 + 6 = 22$. Thus, including the jester cards, there are a total of 22 cards. The player cards will be $22 - 6 = 16$ cards; 16 is a multiple of 8 (for more information, see the 0 or 1 game foundation document). The jester cards are A, B, C, D, E and F.

2 Preliminary

Many engineering systems depend on more than one independent variable. For example, temperature may depend on spatial coordinates, electric potential may vary with position, and mechanical stress may depend on several geometric and physical parameters. Differential calculus of functions of several variables provides mathematical tools for describing and analyzing such systems.

This chapter introduces multivariable functions, limits and continuity, partial derivatives, higher-order derivatives, gradients, directional derivatives, total differentials, tangent planes, the chain rule, implicit differentiation, local extrema, and constrained optimization. These topics are fundamental to engineering applications [13].

2.1 Functions of Several Variables

A function of two variables is a rule

$$f : D \subseteq \mathbb{R}^2 \longrightarrow \mathbb{R}$$

that assigns a unique real number $f(x,y)$ to each point $(x,y) \in D$. Its graph is generally a surface in three-dimensional space. A level curve is a curve defined by

$$f(x,y) = c,$$

where c is a constant [13].

Example. Let

$$f(x,y) = 2x - y^2.$$

At the point $(3,2)$,

$$f(3,2) = 2(3) - 2^2 = 6 - 4 = 2.$$

The level curve corresponding to $c = 4$ is

$$2x - y^2 = 4.$$

2.2 Limits and Continuity

The limit of $f(x,y)$ as (x,y) approaches (a,b) is L if $f(x,y)$ approaches L as (x,y) approaches (a,b) through all possible paths in the domain. The function is continuous at (a,b) if

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b).$$

Limits and continuity are discussed in [9].

Example. Consider

$$f(x,y) = x^2 + 2y^2.$$

Since this is a polynomial, it is continuous everywhere. Therefore,

$$\lim_{(x,y)\to(1,-1)} (x^2 + 2y^2) = 1^2 + 2(-1)^2 = 3.$$

2.3 Partial Derivatives

The partial derivative with respect to one variable measures the instantaneous rate of change with respect to that variable while the other variables are held constant. For a function $f(x,y)$, the first-order partial derivatives are

$$f_x(x,y) = \frac{\partial f}{\partial x}, \qquad f_y(x,y) = \frac{\partial f}{\partial y}.$$

Example. If

$$f(x,y) = 3x^2y + 2y^3,$$

then

$$f_x(x,y) = 6xy, \qquad f_y(x,y) = 3x^2 + 6y^2.$$

2.4 Higher-Order Partial Derivatives

Higher-order partial derivatives are obtained by differentiating partial derivatives again. For example,

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \quad f_{yy} = \frac{\partial^2 f}{\partial y^2}, \quad f_{xy} = \frac{\partial^2 f}{\partial y\, \partial x}.$$

If the relevant second partial derivatives are continuous in a neighborhood, then the mixed partial derivatives are equal:

$$f_{xy} = f_{yx}.$$

Example. Let

$$f(x, y) = x^3 y^2.$$

Then

$$f_x = 3x^2 y^2, \qquad f_{xy} = 6x^2 y,$$

and

$$f_y = 2x^3 y, \qquad f_{yx} = 6x^2 y.$$

Hence $f_{xy} = f_{yx}$.

2.5 Gradient and Directional Derivative

The gradient of a differentiable function $f : \mathbb{R}^2 \to \mathbb{R}$ is the vector

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle .$$

For a unit vector $\mathbf{u} = \langle u_1, u_2 \rangle$, the directional derivative is

$$D_{\mathbf{u}} f(x, y) = \nabla f(x, y) \cdot \mathbf{u}.$$

The gradient points in the direction of the greatest local increase of a differentiable function [8].

Example. Let

$$f(x, y) = x^2 + xy.$$

Then

$$\nabla f(x, y) = \langle 2x + y, x \rangle.$$

At $(1, 2)$,

$$\nabla f(1, 2) = \langle 4, 1 \rangle.$$

2.6 Total Differential and Linear Approximation

For a differentiable function $z = f(x, y)$, the total differential is

$$dz = f_x(x, y)\, dx + f_y(x, y)\, dy.$$

For sufficiently small changes dx and dy, the change in the function can be approximated by

$$\Delta z \approx dz.$$

Example. If

$$f(x, y) = x^2 + 3y,$$

then

$$dz = 2x\,dx + 3\,dy.$$

At $(2, 1)$, this becomes

$$dz = 4\,dx + 3\,dy.$$

2.7 Tangent Plane

For a differentiable surface $z = f(x, y)$, the tangent plane at $(a, b, f(a, b))$ is given by

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

Example. Consider $z = x^2 + y^2$ at $(2, 1, 5)$. Since $f_x = 2x$ and $f_y = 2y$, the tangent plane is

$$z = 5 + 4(x - 2) + 2(y - 1),$$

or equivalently,

$$z = 4x + 2y - 5.$$

2.8 Chain Rule

Suppose $z = f(x, y)$, where $x = x(t)$ and $y = y(t)$. If the functions are differentiable, the chain rule gives

$$\frac{dz}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt}.$$

Example. Let

$$z = x^2 + 3y, \qquad x = 2t, \qquad y = t^2.$$

Substitution gives

$$z = (2t)^2 + 3t^2 = 7t^2.$$

Thus,

$$\frac{dz}{dt} = 14t.$$

2.9 Implicit Differentiation

When variables are related by an equation $F(x, y, z) = 0$, one can differentiate implicitly to find derivatives without first solving explicitly for one variable. If $F_z \neq 0$, then

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}.$$

Example. Suppose

$$x^2 + y^2 + z^2 = 25.$$

Differentiating with respect to x, while holding y constant, gives

$$2x + 2z\frac{\partial z}{\partial x} = 0.$$

Therefore, when $z \neq 0$,

$$\frac{\partial z}{\partial x} = -\frac{x}{z}.$$

2.10 Critical Points and Local Extrema

An interior critical point of a differentiable function $f(x, y)$ is a point where

$$f_x(x, y) = 0, \qquad f_y(x, y) = 0.$$

For a twice-differentiable function, define

$$D = f_{xx} f_{yy} - (f_{xy})^2.$$

At a critical point:

- If $D > 0$ and $f_{xx} > 0$, there is a strict local minimum.
- If $D > 0$ and $f_{xx} < 0$, there is a strict local maximum.
- If $D < 0$, the point is a saddle point.
- If $D = 0$, the test is inconclusive.

This test is described in [13].

Example. Let

$$f(x, y) = x^2 + 4y^2.$$

Since

$$f_x = 2x, \qquad f_y = 8y,$$

the only critical point is $(0, 0)$. Also,

$$D = (2)(8) - 0^2 = 16 > 0, \qquad f_{xx} = 2 > 0.$$

Thus, $(0, 0)$ is a strict local minimum. It is also an absolute minimum because $f(x, y) \geqslant 0$ for all (x, y).

2.11 Lagrange Multipliers

Lagrange multipliers are used to optimize a differentiable objective function $f(x, y)$ subject to a differentiable constraint $g(x, y) = c$. At a regular constrained optimum, one solves

$$\nabla f(x, y) = \lambda \nabla g(x, y), \qquad g(x, y) = c,$$

where λ is a scalar multiplier.

Example. Maximize $f(x, y) = x + y$ subject to

$$x^2 + y^2 = 2.$$

The Lagrange equations are

$$1 = 2\lambda x, \qquad 1 = 2\lambda y, \qquad x^2 + y^2 = 2.$$

They give $x = y = 1$ or $x = y = -1$. Hence the maximum is

$$f(1, 1) = 2.$$

3 Jester and Player Cards

3.1 Instructions

Each question has six answer choices, labelled A, B, C, D, E, and F. Exactly one choice is correct for each question, and the other five are incorrect.

During each play, a student selects one answer choice. A correct selection earns a score of 1, while an incorrect selection earns a score of 0.

3.2 Jester Cards

The jester cards contain the texts A, B, C, D, E and F.

3.3 Player Cards

Question 1. Let $f(x,y) = 2x^2 - y$. What is $f(1,3)$?

(A) -1
(B) 1
(C) 5
(D) -5
(E) 3
(F) 2

Question 2. The level curve $f(x,y) = 4$ for $f(x,y) = x^2 + 4y^2$ is:

(A) $x^2 + y^2 = 4$
(B) $x^2 + 4y^2 = 1$
(C) $x^2 + 4y^2 = 4$
(D) $x + 4y = 4$
(E) $x^2 - 4y^2 = 4$
(F) $4x^2 + y^2 = 4$

Question 3. Evaluate

$$\lim_{(x,y)\to(2,1)} (x^2y + y).$$

(A) 3
(B) 4
(C) 6
(D) 5
(E) 7
(F) 8

Question 4. Let

$$f(x,y) = \frac{x+2y}{x^2+y^2}.$$

What is $f(1,1)$, and is the function continuous at $(1,1)$?

(A) 3, and it is continuous

(B) 1, and it is continuous

(C) $\frac{3}{2}$, and it is continuous

(D) $\frac{1}{2}$, and it is discontinuous

(E) 2, and it is discontinuous

(F) It is undefined

Question 5. If $f(x,y) = xy^3 + 2x^2$, find $f_x(2,3)$.

(A) 4

(B) 6

(C) 8

(D) 10

(E) 35

(F) 14

Question 6. If $f(x,y) = xy^3 + 2x^2$, find $f_y(1,2)$.

(A) 4

(B) 6

(C) 8

(D) 10

(E) 16

(F) 12

Question 7. Let $f(x,y) = x^4y + 2y^3$. Calculate $f_{xy}(1,2)$.

(A) 2

(B) 4

(C) 6

(D) 8

(E) 12

(F) 16

Question 8. Let $f(x,y) = 3x^2 + xy + 2y^2$. Find $\nabla f(1,2)$.

(A) $\langle 6, 8 \rangle$

(B) $\langle 8, 9 \rangle$

(C) $\langle 7, 9 \rangle$

(D) $\langle 8, 8 \rangle$

(E) $\langle 9, 8 \rangle$

(F) $\langle 6, 9 \rangle$

Question 9. For $f(x, y) = x^2 + y^2$, find the directional derivative at $(1, 2)$ in the direction of the unit vector $\mathbf{u} = \langle 3/5, 4/5 \rangle$.

(A) $\dfrac{18}{5}$

(B) $\dfrac{16}{5}$

(C) $\dfrac{22}{5}$

(D) $\dfrac{24}{5}$

(E) 4

(F) 5

Question 10. Let $f(x, y) = \ln x + y^2$. Use the total differential to estimate the change in f at $(1, 2)$ when $dx = 0.02$ and $dy = -0.01$.

(A) 0.06

(B) 0.02

(C) -0.06

(D) -0.02

(E) 0.04

(F) -0.04

Question 11. Find the tangent plane to $z = x^2 + xy$ at $(1, 1, 2)$.

(A) $z = 2x + y - 1$

(B) $z = 3x + y - 2$

(C) $z = 3x + y + 2$

(D) $z = x + 3y - 2$

(E) $z = 3x + 2y - 3$

(F) $z = 2x + 3y - 3$

Question 12. Suppose $z = x^2 y$, where $x = t + 1$ and $y = t^2$. Find $\dfrac{dz}{dt}$ at $t = 1$.

(A) 4

(B) 8
(C) 10
(D) 12
(E) 14
(F) 16

Question 13. If $x^2 + y^2 + z = 12$, find $\dfrac{\partial z}{\partial x}$ at $(x, y) = (2, 1)$.

(A) -2
(B) 2
(C) -1
(D) 4
(E) -4
(F) 0

Question 14. Find the critical point of

$$f(x, y) = x^2 + 4x + y^2 - 6y.$$

(A) $(2, -3)$
(B) $(-2, 3)$
(C) $(-4, 6)$
(D) $(2, 3)$
(E) $(-2, -3)$
(F) $(0, 0)$

Question 15. Classify the critical point $(0, 0)$ of

$$f(x, y) = x^2 - 4y^2.$$

(A) Strict local minimum
(B) Strict local maximum
(C) Absolute minimum
(D) Absolute maximum
(E) Saddle point
(F) Not a critical point

Question 16. Find the maximum value of $f(x, y) = xy$ subject to $x + y = 10$, where $x, y \in \mathbb{R}$.

(A) 10
(B) 20
(C) 24
(D) 25
(E) 50
(F) No maximum exists

4 Answer Key

Question	Answer	Question	Answer
1	A	9	C
2	C	10	D
3	D	11	B
4	C	12	D
5	E	13	E
6	F	14	B
7	B	15	E
8	B	16	D

Scoring rule:

$$\text{Score} = \begin{cases} 1, & \text{if the selected choice is correct,} \\ 0, & \text{if the selected choice is incorrect.} \end{cases}$$

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